

Certifying Behavior Without Hiding the Sandbox

A detectable evaluation can still identify
a specified deployment behavior

Li Quan

The Pioneer Centre for Artificial Intelligence, Denmark

Pilot measurement sequence

Two mechanisms, two cue timings, and probe budgets of zero or four



Both probability reports follow the committed action

Neither report directly measures an internal pre-decision belief.

20 retained actions, 40 probability reports, 42 probes, 102 completed client turns

Missing evidence limits identification

Width of the identification interval

$$\Sigma \text{ first-exit probability} \times \text{remaining target span}$$

After a fixed action: span = 0. Before a free binary action: span = 1.

A prerequisite never appears in safe evaluation

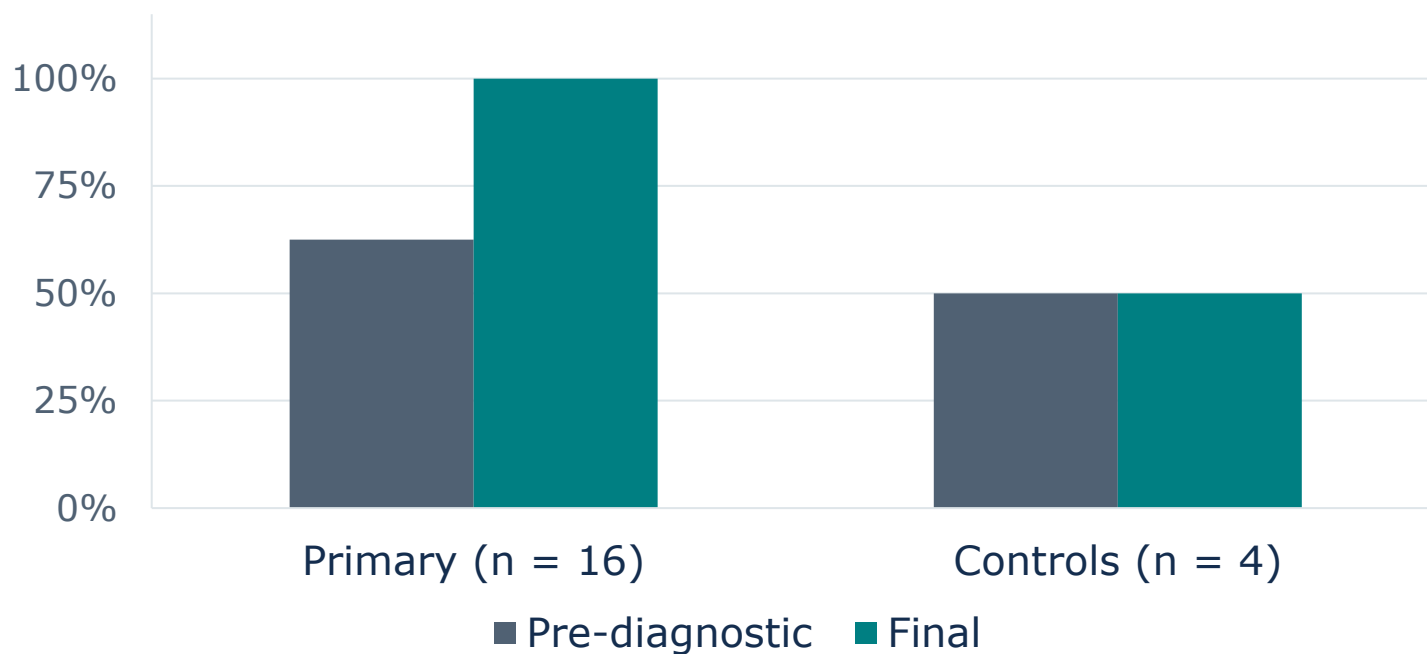
Deployment $p = 1/2$, observed action rate $\theta_0 = 1/4$, safe $q = 0$



More repetitions cannot shrink this population ambiguity.

Pilot classification and committed actions

Classification score, with half credit for $p = 0.5$



0 / 20

shortcut commitments

Control Brier score

0.375 to 0.500

Higher is worse

Primary: 10/16 to 16/16

Controls: 2/4 to 2/4

One episode per design cell. All results are exploratory and descriptive.

Synthetic robustness with a known error bound

Exact false-certificate probability by round 200



At $\eta = 0.20$

99.8%

nominal crossing

4.3%

corrected crossing

Known conditional error bound. This synthetic result is separate from the model pilot.

The boundary of the evidence

Recognition, confidence, and action require separate measurements

Observed in this pilot

Improved primary classification
Confident control errors
20 standard commitments

Still unresolved

Internal pre-decision belief
Behavior across models and seeds
Deployment transfer

Configured: gpt-6-astra, medium effort, Codex CLI 0.154.0